



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiit@gmail.com, website: www.iiitsonepat.ac.in

END TERM

Subject: COMMUNICATION SKILLS (HUL 204)

Branch: CS

M. M. 50marks

Semester II 2022-26 Batch

1221008

Attempt All Questions. All questions carry equal marks.

1. Write an essay on the differences between written and visual communication. Explain gestalt theory of visual communication with the help of suitable examples.

(1 x 10= 10)

2. Read the text in the box carefully. Which type of writing is it? Comment on purpose, nature, effect on its reader and language structure of the text below.

(1 x 10= 10)

"I am sending you a copy of my CV in order to apply for the managerial position which you advertised in the local newspaper. You will find that I have exceptional abilities when it comes to this type of role and my past experience will further demonstrate this.

In my previous post, I successfully managed a team of twenty people whose efforts under my guidance, brought the company into new financial heights.

I am capable and willing to expand my knowledge through training and applied learning. I think you will find that there is no-one more qualified for this position than me."

3. Write a detailed note on non-verbal communication. Discuss its types and explain them with the help of suitable examples.

(1 x 10= 10)

4. Discuss the significance of a thesis statement and topic sentences. Give a suitable example and elaborate on its features.

(1 x 10= 10)

5. Write a detailed note on issues and challenges to comprehension with respect to Language as connected speech.

(1 x 10= 10)



2202

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT
भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत
(An Autonomous Institute of National Importance under Act of Parliament)

Phone: +91 1302987902, Email: sonepatiiit@gmail.com, website: www.iiitsonapat.ac.in

END SEM 2nd

Branch: CSE

Subject Name: WD

M.M.-50 marks

Subject Code: WD (CSC205)

Roll no .12211008

Date:20 JULY 2023

Time:3HRS

Note: All questions are compulsory.

Q-1: (a)	What are the events in Java script? Write the code for any four events in Java script. Why do we use Add Event listener method.	(5 marks)
(b)	What are the looping structure in java script. Write a script for calculating the sum of even integer from 5 to 100 using for loop. Also explain increment and decrement operators in java script.	(5 marks)
Q-2: (a)	How exception handling take place in JavaScript? Discuss with examples.	(5 marks)
(b)	What are the regular expressions? How to create regular expression in java script. Write a script to perform validation by creating a input text field and a submit button.	(5 marks)
Q-3: (a)	How closures work in JavaScript? Discuss with examples. How does strings equals(==) operator differ from equal(===) operator in JavaScript.	(5 marks)
(b)	Write a script that takes four integers from user and displays the sum, average, product, smallest and largest of the numbers in an alert dialog. Also explain the use of conditional operator in java script. Is it equivalent to if else statement?	(5 marks)
Q-4: (a)	Explain the steps to be taken while designing a website. What is white space in web designing.	(5 marks)
(b)	Explain valid and well formed XML document? Distinguish between external and internal DTD declaration.	(5 marks)
Q-5: (a)	Explain <table> tag. Explain the following tags <th>, <td>, <tr>, <tfoot>, <rowspan> and <colspan> with examples.	(5 marks)
(b)	What are the different ways to create an object in java Script? Discuss with examples. Also explain the use of random function in Java Script(1.5 marks)	(5 marks)



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT
भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत
(An Autonomous Institute of National Importance under Act of Parliament)

Phone: +91 1302987902, Email: sonepatiit@gmail.com, website: www.iiitsonepat.ac.in

END SEM

Subject Name: Data Structure

Subject Code: CSC203

Branch: CSE 2nd SEM

M.M.-50 marks

Roll no ...122110006

Time- 03 HRS

Note: All questions are compulsory.

Q-1: (a)	Suppose multidimensional arrays P and Q are declared as P(-2: 2, 2: 22) and Q(1: 8, -5: 5, -10: 5) stored in column major order a) Find the length of each dimension of P and Q. b) The number of elements in P and Q. c) Assuming base address (Q) = 400, W = 4, find the effective indices E1, E2, E3 and address of the element Q[3, 3, 3].	(5 marks)
(b)	Discuss the representation of polynomial of single variable using linked list. Write C functions to add two such polynomials represented by linked list.	(5 marks)
Q-2: (a)	Explain binary search tree (BST) and its operations. Drive the time complexity of BST. Make a binary search tree for the following sequence of numbers, show all steps : 45, 32, 90, 34, 68, 72, 15, 24, 30, 66, 11, 50, 10.	(5 marks)
(b)	Design a stack that supports an additional operation: getMedian(). The getMedian() operation should return the median element in the stack in constant time.	(5 marks)
Q-3: (a)	Insert the following sequence of elements into AVL tree, starting with empty tree 71, 41, 91, 56, 60, 30, 40, 80, 50, 55. Also find the minimum array size to represent this tree.	(5 marks)
(b)	How would you implement a circular queue of integers in C using array. Write a C function to delete element from a circular queue implement it using array.	(5 marks)
Q-4: (a)	Solve the following : a) ((A-(B+C)*D)/(E+F)) [Infix to postfix] b) (A+B)*C-(D-E)↑F [Infix to prefix] c) 752+*415- / - [Evaluate the given postfix expression]	(5 marks)
(b)	Solve the following functions. Assume T(1) = O(1) a) T(n) = 2T(n/2) + c n ³ b) T(n) = 2T(n/2) + n log n c) T(n) = T(n-1) + 1/n d) T(n) = T(n/2) + log n	(5 marks)
Q-5: (a)	Write short notes on: (i) AVL Tree (ii) Big-Oh Notation (iii) Garbage Collection	(5 marks)
(b)	Given below is the subroutine for deletion of a node in a queue. Delete (struct node *list, struct node *first, struct node *last) { Struct node *temp; temp=first; first=first->next; first->prev=NULL; free(first); } There is an error in this code. Modify the code so that it deletes exactly the first node of the queue.	(5 marks)



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiit@gmail.com, website: www.iitsonepat.ac.in

B.Tech. 1st Year (2nd Semester) End Examination, 22nd July 2023.

Subject: Mathematics-II

Subject Code: MAL201

Roll No....12211008

Branch: CSE

Max.Marks.-50

Time- 3 Hours

Note: All questions are compulsory. (Rough sketch compulsory).

Q-1	Change the order of Integration and hence evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dx \, dy$	5 Marks
Q-2	(a) Find by double integration the area enclosed by the curves $y = 2 - x$ and $y^2 = 2(2 - x)$. (b) Find the volume of the solid bounded by the paraboloid $y^2 + z^2 = 4x$ and the plane $x = 5$	2 Marks 3 Marks
Q-3	(a) Find the directional derivative of the divergence of $f(x, y, z) = xy\hat{i} + xy^2\hat{j} + z^2\hat{k}$ at the point (2, 1, 2) in the direction of the outer normal to the sphere $x^2 + y^2 + z^2 = 9$. (b) Let $\vec{v}(x, y, z)$ be a differentiable vector point function and $\phi(x, y, z)$ be a scalar point function. Derive an expression for $\text{div}(\phi\vec{v})$ in terms of $\phi, \vec{v}, \text{div}\vec{v}$ and $\nabla\phi$.	3 Marks 2 Marks
Q-4	Apply Stoke's theorem to evaluate $\int_c 4y \, dx + 2z \, dy + 6y \, dz$ Where c is the curve of intersection of $x^2 + y^2 + z^2 = 6z$ and $z = x + 3$	5 Marks
Q-5	(a) Solve : $y(xy + 2x^2y^3) \, dx + x(xy - x^2y^2) \, dy = 0$ (b) Solve : $xy(1 + xy^2) \frac{dy}{dx} = 1$	2 Marks 3 Marks
Q-6	Using Picard's method obtain a solution upto 5 th approximation of the equation $\frac{dy}{dx} = y + x$ such that $y(0) = 1$. Also compare the result with the exact solution.	5 Marks
Q-7	(a) Solve by the method of variation of parameter $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$ (b) Solve: $\frac{d^4y}{dx^4} - y = \cos x \cosh x$	3 Marks 2 Marks
Q-8	Solve the simultaneous equations $\frac{dx}{dt} + 2y + \sin t = 0$ and $\frac{dy}{dt} - 2x - \cos t = 0$, given that $x=0$ and $y=1$ when $t=0$	5 Marks
Q-9	(a) Find Laplace transform of $f(t) = t^2 e^{-2t} \cos t$ (b) Using Convolution Theorem find inverse Laplace Transform of $\frac{1}{(s^2+1)(s^2+9)}$	2 Marks 3 Marks
Q-10	A particle P of mass 2 grams moves on the x axis and is attracted towards origin O with a force numerically equal to $8x$. If it is initially at rest at $x = 10$, find its position at any subsequent time (using Laplace Transform Technique) assuming (a) no other force acts (b) a damping force numerically equal to 8 times the instantaneous velocity acts.	5 Marks



Roll No. ...12211008...

CSE (Semester- II)

Total marks: 50

(All questions are compulsory)

1. (a) What is the largest Binary number that can be expressed with 12 bits? Determine the equivalent Hexadecimal and Gray code numbers. (3)

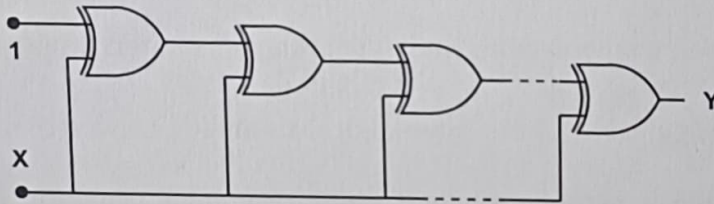
- (b) Given that the following arithmetic operation is correct in at least one number system. Determine the minimum possible base.

$$\frac{302}{20} = 12.1$$

(2)

2. (a) Differentiate between De-Morgan's theorem and Duality theorem with an example. (3)

- (b) What is the output Y if the input to the digital circuit consisting of a cascade of 30 Ex-OR gates is X as shown in figure below. (2)



3. (a) Find the number of prime implicants, essential prime implicants, and the minimized boolean expression for the following function

$$F(A, B, C, D) = \sum_m(0, 2, 5, 9, 15) + \sum_d(6, 7, 8, 10, 12, 13)$$

(3)

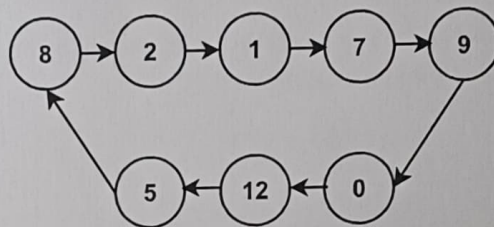
- (b) What is the minimum number of NAND gates required to implement the following boolean function ?

$$A + A\bar{B} + A\bar{B}C$$

(2)

4. Explain 2-bit Comparator with its circuit diagram in details. Write its truth table and find all possible boolean expressions using K-map. (5)

5. (a) Design a boolean function $F(A, B, C) = A\bar{C} + \bar{B}C$ using 3×8 decoder with active low outputs. (3)
- (b) Design a boolean function $F(A, B, C) = \sum_m(0, 2, 5, 7)$ using 4×1 MUX with B (MSB) and C (LSB) selection lines. (2)
6. (a) Construct a T flip-flop from a $J - K$ flip-flop. (3)
- (b) The initial state of MOD-16 asynchronous down counter is 0110. What is the state of the counter after 37 clock pulses. (2)
7. Design a synchronous counter using $J - K$ flip-flops with the following sequence as shown in figure below. (5)



8. (a) Define race-around problem in flip-flops and Prove that the next state equation $Q_s(n+1)$ for a slave flip-flop in master-slave flip-flop is equal to master flip-flop output $Q_m(n)$. (3)
- (b) Differentiate between asynchronous and synchronous counters with an example. (2)
9. (a) Explain the 4-bit weighted resistor type digital-to-analog converter in details. (3)
- (b) Explain the following general terms for a digital-to-analog converter
- Resolution
 - Full scale output
- (2)
10. (a) Explain the following terms
- Memory word
 - Volatile memory
 - Random-access memory (RAM)
- (3)
- (b) A certain semiconductor memory chip is specified as $2K \times 8$. How many words can be stored on this chip? What is the word size? How many total bits can this chip store? (2)

End of question paper